Aarjav Sharma

Security Engineer (Automation and DevOps)

+91 7827850349 | aarjavsharma82@gmail.com | LinkedIn | Github: Aarjav-2003 | LeetCode: aarjavsharma82

Education

VIT Bhopal University 2022 – 2026
B.Tech in Computer Science (Specialization in Cyber Security and Digital Forensics) | CGPA: 8.64/10.0 *Bhopal, India*

Technical Skills

Programming & Scripting: C/C++, Python, Bash
Cloud & DevOps: AWS (EC2, IAM, S3), Jenkins, Git, Ansible, Terraform, Docker
Operating Systems: Linux, Windows

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, Nessus
Frameworks & Databases: Django, MySQL, MongoD, DynamoDB
Core Knowledge: Pentesting, Cryptography, OWASP Top 10 Vulnerability Management, Ethical Hacking

Key Projects

PPA (Predicative Posture Analyst) | NetworkX, JSON, Linux, Graph Theory, Security Analytics 2025

- Architected a graph-theoretic security posture analysis framework, leveraging bash-driven telemetry collection and Python-based data normalization into structured JSON.
- Engineered a dynamic risk graph with NetworkX to model privilege relationships and systemic dependencies, enabling adversarial path prediction.
- Implemented attack path simulation with probabilistic risk scoring, quantifying exploitability and prioritizing remediation across critical assets.
- Automated vulnerability intelligence reporting, consolidating attack trajectories, exposure scores, and actionable insights for analysts.
- Outlined extensibility for web-based visualization dashboards, context-aware mitigation recommendations, and controlled auto-remediation workflows.

Process Killer | Bash, Python, Linux Automation, Email Alerts 2025

- Implemented adaptive process termination, privilege escalation monitoring, and dynamic quarantine with automated alerting workflows.
- Architected and deployed an automated Linux monitoring daemon using Bash for real-time threat mitigation via behavioral heuristics and resource thresholding.
- Integrated Python-driven forensic log analysis module for anomaly detection, event correlation, and incident pattern reporting.

AASA (Fake Banking APK Detector) | Python, Bash, YAML, JSON, Shell Scripting 2025

- Developed an automated framework to detect malicious Android banking applications using static analysis and custom rule-based detection.
- Automated threat intelligence correlation using malicious hash feeds, domain whitelisting, and entropy-based obfuscation scoring to flag high-risk applications.
- Designed domain whitelisting system to detect suspicious network connections while reducing false positives.
- Orchestrated end-to-end pipeline with Bash automation and CLI utilities for scalable APK ingestion, risk scoring, and forensic-grade reporting.

Extracurricular

Co-Lead in Media Team of Freelancing Club, driving digital initiatives and collaborating on outreach campaigns with fellow club members.

Content Team member in BashCraft Club, contributing technical articles, event coverage, and supporting club media strategy.

Participant in major hackathons including Societe Generale, GrabHack and Adobe Hackathons and, gaining exposure to real-world tech problem-solving.

Violinist performing at college events, actively contributing to campus cultural life through music.

Certificates

- CEHv12 by EC-Council
- Cyber Security Analyst by IBM
- Ethical Hacking by NPTEL (Silver Level)
- Bits and Bytes of Computer Networking by Coursera